

Project Design Phase

Problem – Solution Fit Template

Date	15 July 2026
Team ID	
Project Name	UCab Premium – Cab Booking Application
Maximum Marks	2 Marks

The Problem–Solution Fit below shows how UCab Premium addresses the real problems and behaviors of riders and drivers.

1. Customer Segment(s)	6. Customer Constraints	5. Available Solutions
Urban commuters needing on-demand transport (Riders); part-time and full-time cab drivers seeking ride requests (Drivers); platform operators (Admin).	Smartphone/browser access required; dependence on mobile network connectivity; budget-conscious riders comparing fares.	Traditional taxi stands, phone-based dispatch, and other ride-hailing apps; most lack transparent, real-time status tracking for both rider and driver.

2. Jobs-to-be-Done / Problems	9. Problem Root Cause	7. Behaviour
Riders need to book a trustworthy, fairly-priced cab quickly. Drivers need a steady, organized stream of ride requests.	No centralized digital platform connecting riders and drivers with real-time status and role-based accountability; dispatch is manual and informal.	Riders currently compare multiple apps or call drivers directly; drivers wait for word-of-mouth or informal calls for ride requests.

3. Triggers	10. Your Solution	8. Channels of Behaviour
Need to travel for work/personal reasons; a driver's shift starting with no assigned rides; frustration after a bad experience with an unreliable ride.	UCab Premium — a MERN-stack platform with role-based Rider, Driver, and Admin portals, JWT authentication, a five-stage booking lifecycle, transparent fare display, PDF receipts, and analytics dashboards.	Online: Web app (React + Vite) accessible via browser, with REST APIs (Node.js/Express) powering booking, assignment, and payment flows.

4. Emotions: Before / After	Description
Before: anxious, uncertain about wait times and driver assignment. After: confident, reassured by real-time status and a rated, verified driver.	The booking lifecycle (Pending → Driver Assigned → Accepted → On the Way → Completed) removes uncertainty at every step, directly improving the rider's and driver's emotional experience.